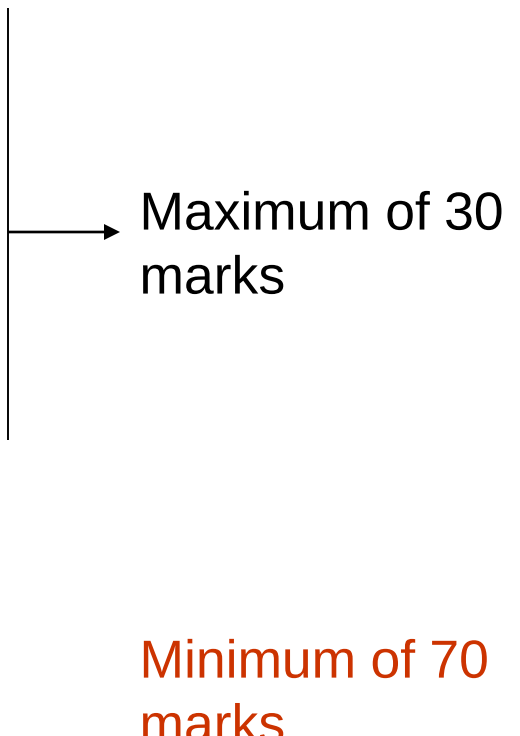


CHE 2105

Chemistry of s,p,d elements and
Nuclear Chemistry& its
applications

Mode of Evaluations

- Quizzes
 - Tutorials
 - Spot test
 - Mid-semester examinations
- 
- Maximum of 30 marks
- Final examinations → Minimum of 70 marks

Final Examinations

- Answer both questions.
- Question no. 01 – ten parts (a), (b), .. (j) each part carry 12 marks and $10 \times 12 = 120$ marks.
- Question no.02 – has part (A) and part (B)
- Each question has 3-4 parts and carry 80 marks each.

Chemistry of s,p,d elements

- References:

- Concise Inorganic Chemistry 5th ed. J.D.Lee
- Chemistry Raymond Chang
- Inorganic Chemistry Shriver and Atkins

Objectives

- To study the properties of the elements and their compounds.

Gro up	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Peri od 1	¹ H																	² He
2	³ Li	⁴ Be											⁵ B	⁶ C	⁷ N	⁸ O	⁹ F	¹⁰ Ne
3	¹¹ Na	¹² Mg											¹³ Al	¹⁴ Si	¹⁵ P	¹⁶ S	¹⁷ Cl	¹⁸ Ar
4	¹⁹ K	²⁰ Ca	²¹ Sc	²² Ti	²³ V	²⁴ Cr	²⁵ Mn	²⁶ Fe	²⁷ Co	²⁸ Ni	²⁹ Cu	³⁰ Zn	³¹ Ga	³² Ge	³³ As	³⁴ Se	³⁵ Br	³⁶ Kr
5	³⁷ Rb	³⁸ Sr	³⁹ Y	⁴⁰ Zr	⁴¹ Nb	⁴² Mo	⁴³ Tc	⁴⁴ Ru	⁴⁵ Rh	⁴⁶ Pd	⁴⁷ Ag	⁴⁸ Cd	⁴⁹ In	⁵⁰ Sn	⁵¹ Sb	⁵² Te	⁵³ I	⁵⁴ Xe
6	⁵⁵ Cs	⁵⁶ Ba	⁵⁷ La	⁷² Hf	⁷³ Ta	⁷⁴ W	⁷⁵ Re	⁷⁶ Os	⁷⁷ Ir	⁷⁸ Pt	⁷⁹ Au	⁸⁰ Hg	⁸¹ Tl	⁸² Pb	⁸³ Bi	⁸⁴ Po	⁸⁵ At	⁸⁶ Rn
7	⁸⁷ Fr	⁸⁸ Ra	⁸⁹ Ac	¹⁰⁴ Rf	¹⁰⁵ Db	¹⁰⁶ Sg	¹⁰⁷ Bh	¹⁰⁸ Hs	¹⁰⁹ Mt	¹¹⁰ Ds	¹¹¹ Rg							
			⁵⁸ Ce	⁵⁹ Pr	⁶⁰ Nd	⁶¹ Pm	⁶² Sm	⁶³ Eu	⁶⁴ Gd	⁶⁵ Tb	⁶⁶ Dy	⁶⁷ Ho	⁶⁸ Er	⁶⁹ Tm	⁷⁰ Yb	⁷¹ Lu		
			⁹⁰ Th	⁹¹ Pa	⁹² U	⁹³ Np	⁹⁴ Pu	⁹⁵ Am	⁹⁶ Cm	⁹⁷ Bk	⁹⁸ Cf	⁹⁹ Es	¹⁰⁰ Fm	¹⁰¹ Md	¹⁰² No	¹⁰³ Lr		

- Most of the elements in nature are in elemental form or in chemical combination with other elements.
- Most of these elements are stable but some of these elements (heavy elements) are radioactive.
- Some of these elements have extremely short lifetimes. And can be obtained synthetically. I.e obtained by nuclear reactions.
- The last few elements in the periodic table have only been produced in trace amounts and decay very rapidly into lighter elements.

Why do chemists study elements and their compounds.

- The use of metals such as Fe (iron), Cu (copper), Al (aluminium), Sn (tin), Pb (lead), Zn (zinc), Ag (silver) and Au (gold) for human beings in many ways are well known.
- Other metals and their compounds are also important. For example V_2O_5 is used in the manufacture of H_2SO_4

The modern periodic Table (based on the electronic configurations of the elements)

- We can identify four main types.
 - Noble Gases (Gp viii or 18 elements)
 - Representative elements (non transition elements)
 - Transition elements (filling of d orbitals)
 - Inner transition elements (filling of f orbitals)

The properties of the elements in the periodic Table could be discussed under the following categories.

- Hydrogen and noble gases.
- The alkali and alkaline earth metals (s-block)
- Non metals and metalloids (p block)
- Transition elements (d block)

Hydrogen and noble gases.

- The properties of hydrogen and its compounds have many unique properties.
- The noble gas elements also have some unique properties which are associated with completely filled s and p orbitals.
- Note : Noble gas elements are sometimes incorrectly referred to as inert gas elements.

The alkali and alkaline earth metals (s-block)

- Alkali metals
 - Li
 - Na
 - K
 - Rb(Rubidium)
 - Cs(Cesium)
 - Fr(Francium)

Alkaline earth metals

- Be
- Mg
- Ca
- Sr(strontium)
- Ba
- Ra (radium

Non metals and metalloids (p block)

- Gp III – B,Al Ga,In Tl (Thallium) – ns^2np^1
- Gp IV – C,Si, Ge, Sn Pb – ns^2np^2
- GP V – N,P,As,Sb(antimony), Bi – ns^2np^3
- GP VI – O,S,Se, Te (tellurium), Po (polonium) – ns^2np^4
- Gp VII – (the halogens) F,Cl,Br,I, At (Astatine) - ns^2sp^5

Transition elements (d block)

- Elements formed by filling the 3d, 4d and 5d shells of electrons.
- They are often called transition because their position in the periodic table is between the s-block and p-block elements.
- i.e. Their properties are transitional between the highly reactive metallic elements (s-block) and the elements of the p-block.

In respect of each category we shall discuss the following ;

- Electronic configurations of the atoms.
- Physical and Chemical properties of the elements and some of their important compounds.

The Chemistry of Hydrogen

- Hydrogen having the simplest atomic structure consisting of one p (proton) and one e (electron)
- The electronic structure is $1s^1$

- Hydrogen combines with every element in the periodic table except the nonmetals in Group VIIIA (He, Ne, Ar, Kr, Xe, and Rn).
- Most carbon compounds also contain hydrogen, and hydrogen forms compounds with virtually all the other elements as well.
- Compounds of hydrogen are frequently called **hydrides**, even though the name hydride literally describes compounds that contain an H-ion.
- This trend is so regular that the combining power, or *valence*, of an element was once defined as the number of hydrogen atoms bound to the element in its hydride.

- Hydrogen is the only element that forms compounds in which the valence electrons are in the $n = 1$ shell.
- As a result, hydrogen can have three oxidation states, corresponding to
 - the H^+ ion,
 - a neutral H atom,
 - and the H^- ion.

$$\text{H}^+ = 1s^0$$

$$\text{H} = 1s^1$$

$$\text{H}^- = 1s^2$$

Hydrogen's Placement in the Periodic Table

- Like that of the alkali metals.

The alkali metals also have just one electron in their outer shell.

H^+ are known.

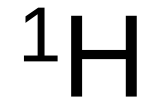
- Like that of halogens (Gp VII)
- Both are one electron short of a noble gas structure.
- In many reactions the halogens gain an electron and form negative ions X^- .
- Hydrogen does form ionic hydrides M^+H^-
e.g., LiH and CaH_2

- Hydrogen resembles that of the Gp IV elements, since both have half filled shell of electron.
- There are number of similarities between hydrides and organometallic compounds. CH_3^- and H^- both have one remaining valency.
- Hydride is often considered as part of a series of organometallic compounds.
- For example LiH , LiCH_3 , LiEt ; NH_3 , NMe_3 , NEt_3 etc.,

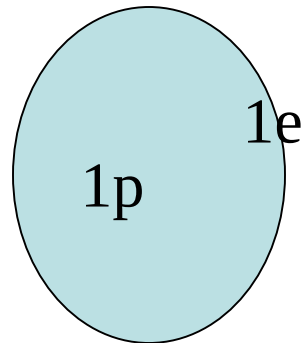
- Therefore there is no single completely satisfactory place in the periodic table that can be assigned to hydrogen.
- Hydrogen is best treated as a group on its own.

Isotopes of Hydrogen

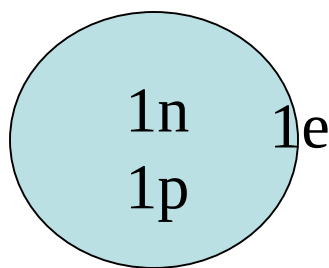
- Three isotopes of hydrogen are known
 1. ^1H
 2. ^2H (Deuterium)
 3. ^3H (Tritium)



- 1 proton and 1 electron



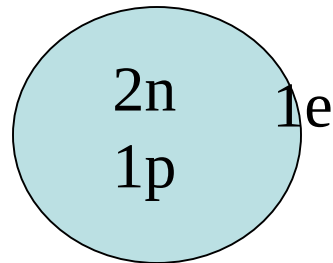
^2H (Deuterium)



1 proton, 1 neutron and 1 electron

^3H (Tritium)

- 1 proton, 2 neutrons and 1 electron



- Most chemical properties of H, D and T are identical. Naturally occurring hydrogen contains 0.015% D.
- Radioactive Tritium is formed in nuclear fusion reactions.

Formation of tritium

- Natural formation of tritium in the upper atmosphere is due to nuclear reactions induced by cosmic rays.

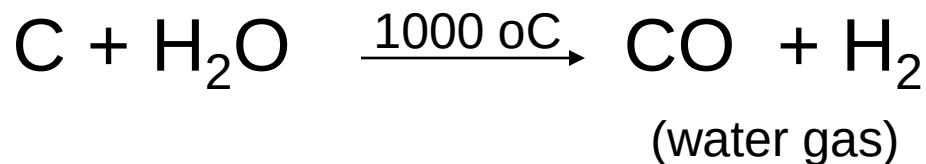
Eg. Hydrogen bomb

- Fusion of these isotopes of hydrogen ultimately forms stable helium with the liberation of large amount of energy.
- ${}^2\text{H} + {}^3\text{H} \longrightarrow {}_2^4\text{H} + {}^1_0\text{n} + \text{energy}$
- The energies of the sun including stars is supposed to originate from these reactions.

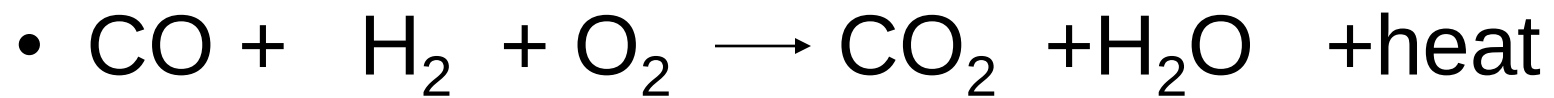


Preparation of Hydrogen

- Hydrogen is manufactured on a large scale by a variety of methods:
- Hydrogen is made cheaply and large amounts by passing steam over red hot coke.



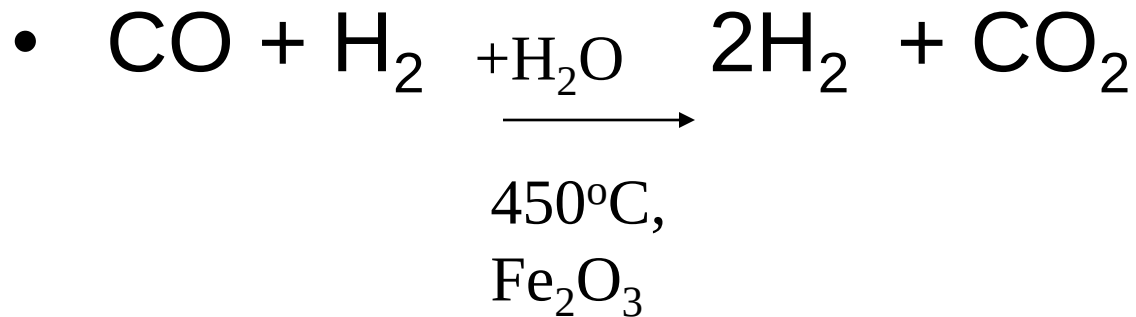
water gas is a very important industrial fuel, since it is easy to make and it burns and evolving a lot of heat.



H₂ from water gas

- It is difficult to obtain pure H₂ from water gas.
- CO is difficult to remove.
- CO may be liquefied at a low temperature under pressure thus separating it from H₂

- Alternatively the gas mixture can be mixed with steam, cooled to 400°C and passed over Fe₂O₃ in a shift converter.



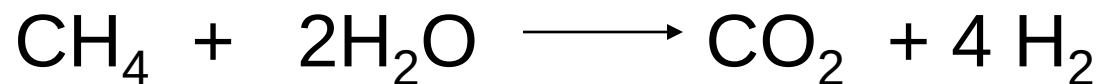
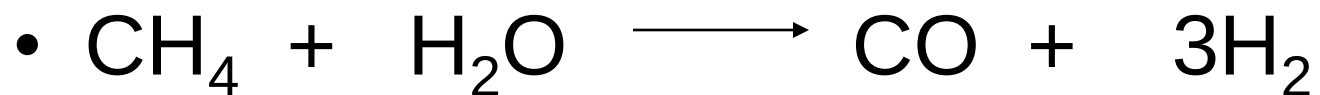
- The CO_2 is easily removed either;
 - by dissolving in water under pressure or
 - reacting with K_2CO_3 solution, giving KHCO_3

2. Steam reformer process

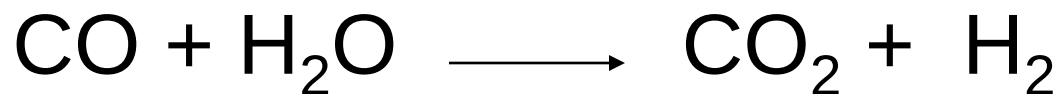
2. Also made in large amounts by the steam reformer process.

Light hydrocarbons (methane) are mixed with steam and passed over a Ni catalyst at 800 – 900°C.

(These hydrocarbons are present in natural gas and are also produced at oil refineries when cracking)



- Mixture of CO, CO₂ and H₂ is mixed with more steam , cooled to 400°C and passed over Fe/Cu catalyst and CO is converted into CO₂.



- Then CO_2 is absorbed in a solution of K_2CO_3 or ethanolamine $\text{HOCH}_2\text{CH}_2\text{NH}_2$
- $\text{K}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow 2\text{KHCO}_3$
- $2\text{HOCH}_2\text{CH}_2\text{NH}_2 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow (\text{HOCH}_2\text{CH}_2\text{NH}_3)_2\text{CO}_3$

3. Cracking of oil

- In oil refineries, natural hydrocarbon mixtures of high (naphtha) and lower molecular weight (petrol) products are produced . Hydrogen is a valuable by-product.

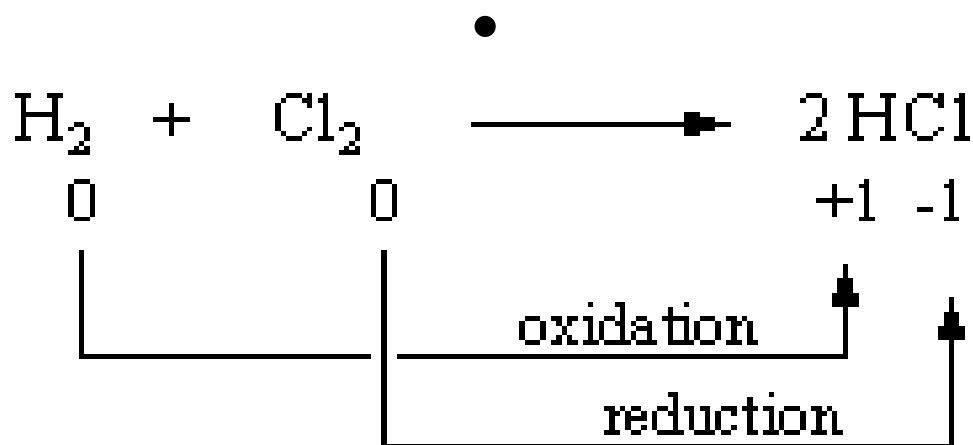
- Because hydrogen forms compounds with oxidation numbers of both +1 and -1, many periodic tables include this element in both Group IA (with Li, Na, K, Rb, Cs, and Fr) and Group VIIA (with F, Cl, Br, I, and At).

- There are many reasons for including hydrogen among the elements in Group IA.
- It forms compounds (such as HCl and HNO_3) that are analogs of alkali metal compounds (such as NaCl and KNO_3). Under conditions of very high pressure, it has the properties of a metal. (It has been argued, for example, that any hydrogen present at the center of the planet Jupiter is likely to be a metallic solid.)
- Finally, hydrogen combines with a handful of metals, such as scandium, titanium, chromium, nickel, or palladium, to form materials that behave as if they were alloys of two metals.

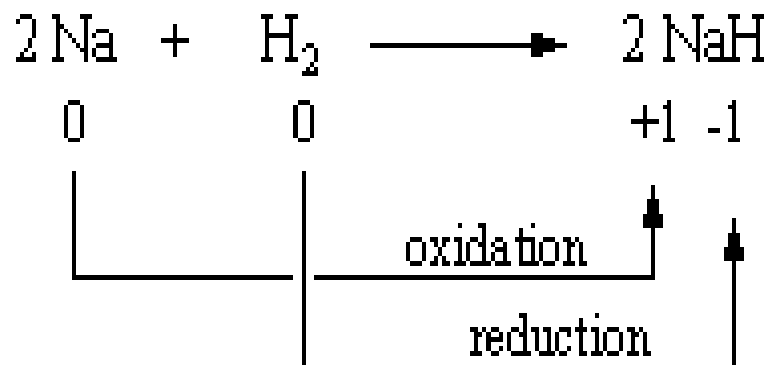
- There are equally valid arguments for placing hydrogen in Group VIIA.
- It forms compounds (such as NaH and CaH_2) that are analogs of halogen compounds (such as NaF and CaCl_2).
- It also combines with other nonmetals to form covalent compounds (such as H_2O , CH_4 , and NH_3), the way a nonmetal should.
- Finally, the element is a gas at room temperature and atmospheric pressure, like other nonmetals (such as O_2 and N_2).

- It is difficult to decide where hydrogen belongs in the periodic table
- Because of the physical properties of the element. The first ionization energy of hydrogen (1312 kJ/mol), for example, is roughly halfway between the elements with the largest (2372 kJ/mol) and smallest (376 kJ/mol) ionization energies.
- Hydrogen also has an electronegativity ($EN = 2.20$) halfway between the extremes of the most electronegative ($EN = 3.98$) and least electronegative ($EN = 0.7$) elements. On the basis of electronegativity, it is tempting to classify hydrogen as a semimetal

- Hydrogen is oxidized by elements that are more electronegative to form compounds in which it has an oxidation number of +1.



- Hydrogen is reduced by elements that are less electronegative to form compounds in which its oxidation number is -1.



Properties and Formation of Hydrogen

- At room temperature, hydrogen is a colorless, odorless gas with a density only one-fourteenth the density of air. Small quantities of H_2 gas can be prepared in several ways.

- 1. By reacting an active metal with water.
- $\text{Na(s)} + 2 \text{H}_2\text{O(l)} \rightarrow 2 \text{Na}^+(\text{aq}) + 2 \text{OH}^-(\text{aq}) + \mathbf{H_2(g)}$

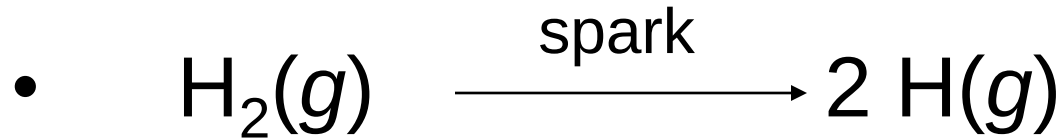
- 2. By reacting a less active metal with a strong acid.
- $\text{Zn(s)} + 2 \text{HCl(aq)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2 \text{Cl}^{-}(\text{aq}) + \mathbf{H_2(g)}$

- 3. By reacting an ionic metal hydride with water.
- $\text{NaH(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq}) + \text{H}_2(\text{g})$

- 4. By decomposing water into its elements with an electric current.



- The covalent radius of a neutral hydrogen atom is 0.0371 nm, smaller than that of any other element.
- Because small atoms can come very close to each other, they tend to form strong covalent bonds.
- As a result, the bond dissociation enthalpy for the H-H bond is relatively large (435 kJ/mol).
- H₂ therefore tends to be unreactive at room temperature. In the presence of a spark, however, a fraction of the H₂ molecules dissociate to form hydrogen atoms that are highly reactive



- The heat given off when H atoms react with O₂ is enough to catalyze the dissociation of additional H₂ molecules.
- Mixtures of H₂ and O₂ that are infinitely stable at room temperature therefore explode in the presence of a spark or flame.

Uses for Hydrogen

- Hydrogen is the lightest element and can be used as a lifting agent in balloons.
- Since Hydrogen gas is highly inflammable, it can be dangerous to use.
- Hydrogen is suitable for use as a fuel.
- The most common use of Hydrogen is in chemical processes and reactions

Lifting agent

- Since the hydrogen atom has an atomic weight of **1** and the oxygen atom has an atomic weight of **16**, you can see that hydrogen is much lighter than air.
- A balloon filled with hydrogen gas (H_2) will readily float upward in air. It seems to be a prime candidate for a lighter-than-air airship, but it also proved to be too dangerous.

Fuel

- Hydrogen can be used as a fuel.

- ***Rockets***

- Since hydrogen is highly inflammable, especially when mixed with pure oxygen, it is used as a fuel in rockets. Usually, they combine liquid hydrogen with liquid oxygen to make an explosive mixture.
- Unfortunately, in 1986, the U.S. Space Shuttle Challenger exploded when a flame accidentally ignited the liquid hydrogen in an external fuel tank. This again showed that the gas can be dangerous and cause a disaster in some situations.

Clean fuel for cars

Hydrogen is one of the cleanest fuels because when it burns, the result is simple water.

That is why there are efforts to create engines that can power automobiles on hydrogen.

This would greatly help to reduce the air pollution and global warming problems.

Although hydrogen is highly inflammable, so also is gasoline. Although care must be taken, the amount of hydrogen used in an automobile would present no more of a hazard than the amount of gasoline used.

One problem with using hydrogen to directly power an automobile is that it is very expensive to create pure hydrogen for this use. Also, the most common method to create hydrogen for use in cars is with methane (**CH₄**) gas. Although burning the hydrogen is pollution-free, methane is a major contributor to the greenhouse effect of global warming.

Thermonuclear energy

- When hydrogen is heated to extreme temperatures, such as on the Sun, the nuclei of hydrogen atoms will fuse to create helium nuclei. This fusion results in the release of an enormous amount of energy, called thermonuclear energy. This process is what creates the energy of the sun.
- By using an Atomic Bomb to create a temperature similar to that on the Sun, thermonuclear fusion was created on the Earth in the form of a Hydrogen Bomb. This bomb was many times more powerful than an Atomic Bomb.
- Scientists are working on harnessing this energy to be used for peaceful means of creating inexpensive energy, especially for generating electrical power. Unfortunately, they have not been able to do this in a usable manner.

- **Chemical processes**
- Hydrogen is an essential part of many chemical processes.

- ***Acids and bases***

- Hydrogen atoms are in every acid and base. Hydrogen gas is sometimes used directly to create an acid.
- For example, it is used in the creation of hydrochloric acid:
- **$\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$.**

- ***Petroleum***

- Hydrogen gas is used in the processing of petroleum products to break down crude oil into fuel oil, gasoline and etc..

- ***Fertilizer***

- Hydrogen is important in creating ammonia (NH_3) for use in making fertilizer.

- ***Food and fat***

- Hydrogen gas is used as a hydrogenating agent to for polyunsaturated fats, such as used in margarine. But it is also used in making unhealthy trans-fats, that are often used in cookies and other goods.

- Note
- Hydrogen was once used in zeppelins and blimps, but it proved too dangerous.
- Since Hydrogen gas is highly inflammable, it can be suitable for use as a fuel. The most common use of Hydrogen is in chemical processes and reactions.

The Chemistry of Water

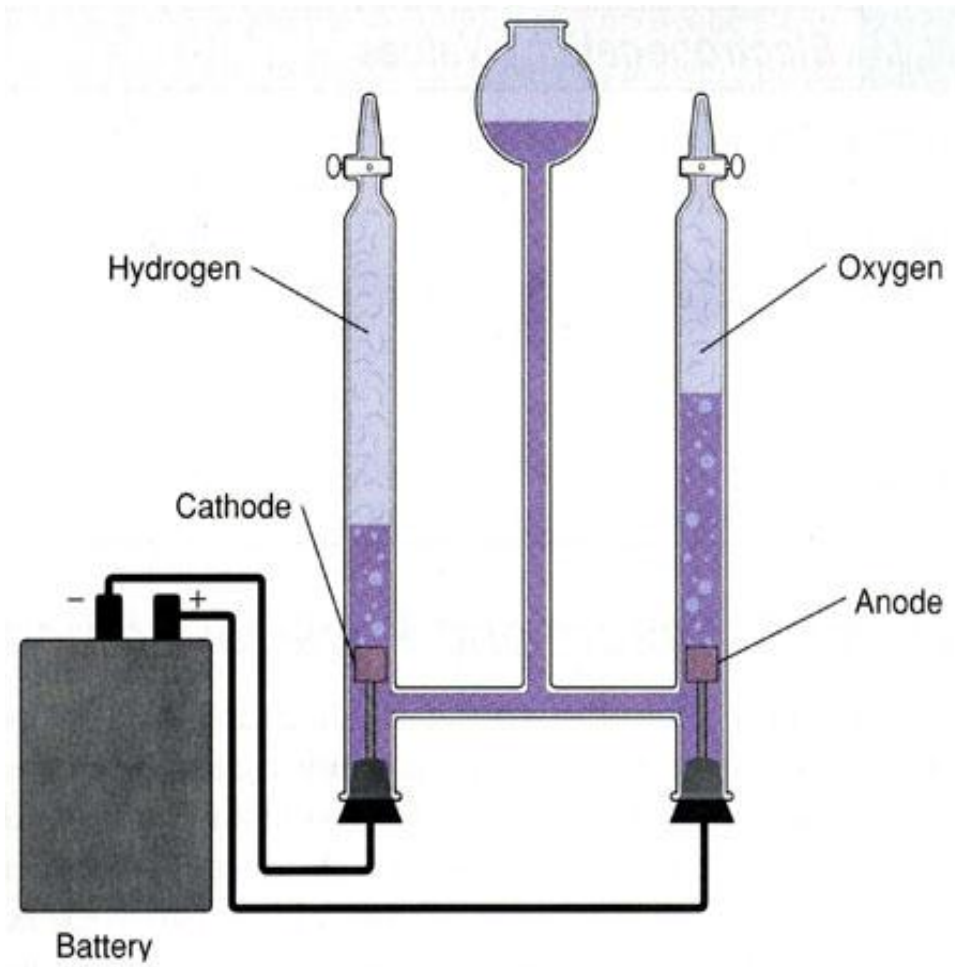
- $2\text{H}_2 + \text{O}_2 = 2\text{H}_2\text{O}$
- It takes two molecules of the diatomic hydrogen gas, combined with one molecule of the diatomic oxygen gas to produce two molecules of water.
- The formation of water from its elements produces, in addition to water, a tremendous amount of energy, 572 kJ to be exact.

- **$2\text{H}_2 + \text{O}_2 = 2\text{H}_2\text{O} + \text{ENERGY}$**
- **This is an example of an exothermic reaction, a reaction that produces energy. It is also an example of what is called a combustion reaction, where a substance (in this case hydrogen gas) is combined with oxygen.**

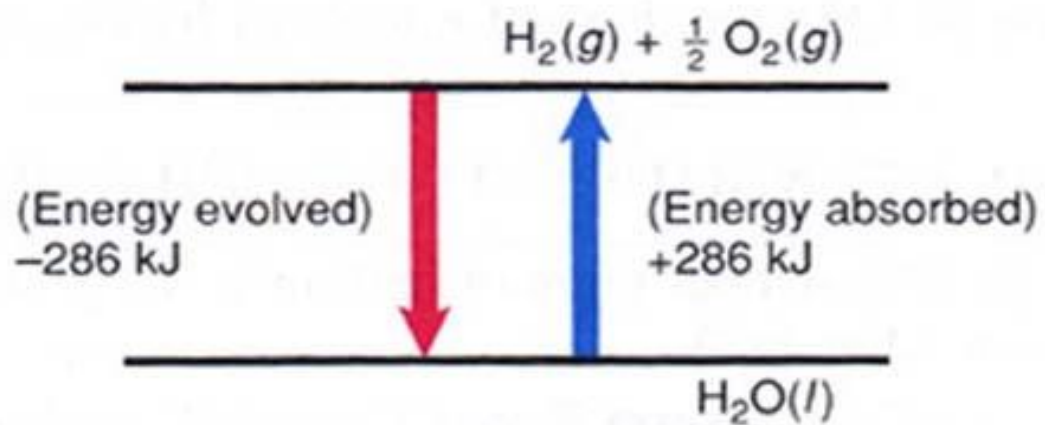
- Hydrogen is a good, clean fuel, producing only water as a by-product.
- Unfortunately it produces so much energy that it can get out of control, resulting in an explosion.
- But let's forget about that explosive part for a minute and think about the possibilities.

- **$2\text{H}_2\text{O} + \text{ENERGY} = 2\text{H}_2 + \text{O}_2$**
- This is an example of an Endothermic reaction.
- IF we could find an easy way to convert the water to hydrogen and oxygen, then the hydrogen could be used as a clean fuel.

- One way to convert Water to Hydrogen and Oxygen is through the process of Electrolysis.



- Using Energy to break water to form hydrogen to combine oxygen to form Energy



Chemistry of O-group elements

- Inert gas or noble gas, any of the elements in Group 18 of the periodic table.
- In order of increasing atomic number they are: helium, neon, argon, krypton, xenon, and radon. They are colorless, odorless, tasteless gases.
- Once believed to be entirely inert, i.e., forming no chemical compounds;
- however, some compounds of these elements have been produced, i.e., fluorides of krypton, xenon, and radon.

- The low chemical activity of the inert gases is
- Due to the fact that their outermost, or valence, electron shell is complete, containing two electrons in the case of helium and eight in the remaining cases. natural gas and the atmosphere.

- The inert gases are sometimes called the rare gases, although argon is not rare (it makes up about 1% of the atmosphere).

1	2											3	4	5	6	7	0	
		H																He
Li	Be											B	C	N	O	F	Ne	
Na	Mg											Al	Si	P	S	Cl	Ar	
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr	
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn	
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg								